NMAP to Discover Live Hosts Using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform.

Exp.no : 8a
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Aim:

To discover live hosts using Nmap through ARP, ICMP, TCP and UDP scanning techniques on the TryHackMe platform.

procedure:

1.) Open the TryHackMe Nmap Live Host Discovery room.
2.) Study the network structure and identify the targets IP addresses or subnets.
3) Specify the targets using an IP address, range, list, or subnets
4.) Use ARP scanning to identify live hosts on the same subnets.
5.) Use ICMP Echo, Timestamp and address mask requests for host discovery.
6.) Use TCP SYN ping and TCP ACK ping to check whether the target hosts are active.
7.) Use UDP ping to identify hosts through UDP and ICMP responses.

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8.) Use -sn when only live host discovery is required without port scanning.

9.) Use reverse DNS lookup when hostnames of discovered systems are required.

10.) Record the responses and identify the live hosts.

Important commands

→ nmap -PR -sn TARGETS - ARP host discovery
→ nmap -PE -sn TARGETS - ICMP Echo discovery
→ nmap -pp -sn TARGETS - ICMP Timestamp discovery
→ nmap -PM -sn TARGETS - ICMP Address Mask discovery.
→ nmap -PS 23, TARGETS - Tcp SYN ping on Telnet port 23
→ nmap -PA TARGETS - TCP ACK ping
→ nmap -R TARGETS - Reverse DNS lookup
→ nmap -n TARGETS - Disable DNS resolution.

ARP is useful for hosts on the same subnet, while ICMP, Tcp and UDP methods can be used for host discovery in different network situations.

Output:

The following observations were obtained during the experiments.

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1.) ARP Request was used to identify active hosts on the same subnet.

2.) ARP Response indicated that the target host was active

3.) ICMP Echo Request and Echo Reply were used for ping-based discovery

4.) ICMP Timestamp discovery used -PP.

5.) ICMP Address Mask discovery Used -PM.

6.) ICMP Echo discovery Used -PE.

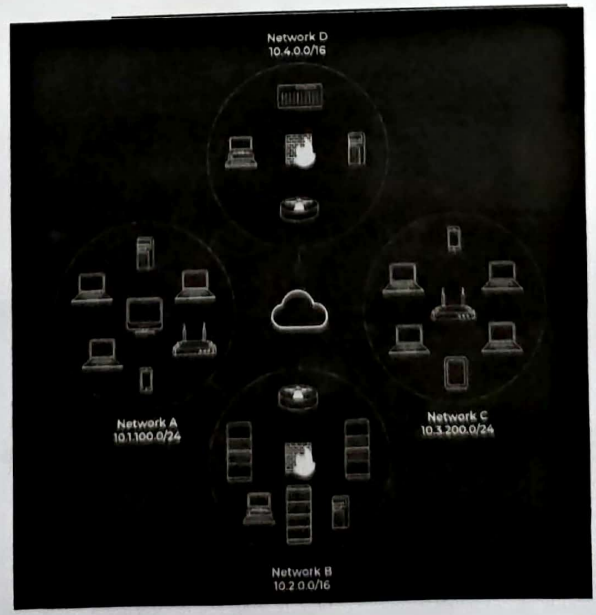
7.) Tcp SYN ping Used -PS

8.) TCP ACK ping Used -PA

9.) UDP ping could identify a host through an ICMP port unreachable response.

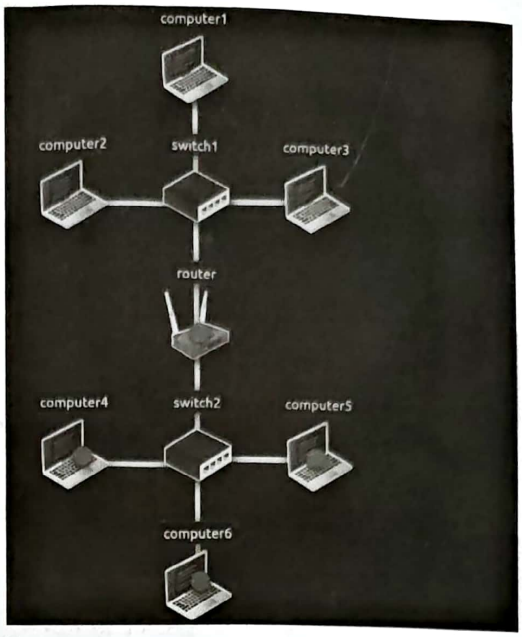
10.) Reverse DNS lookup used -R

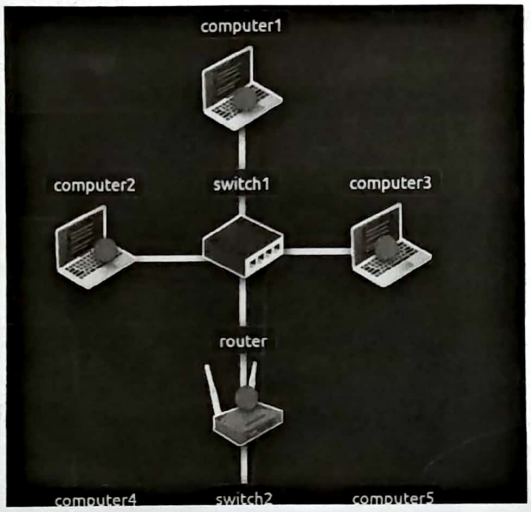
11.) The -sn option was used for host discovery without port scanning.



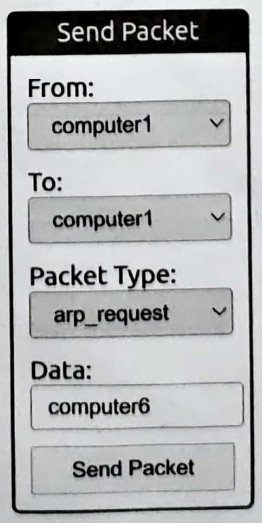
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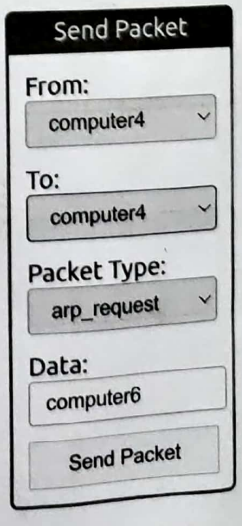
1. Enumerate targets
2. Discover live hosts
3. Reverse-DNS lookup
4. Scan ports
5. Detect versions
6. Detect OS
7. Traceroute
8. Scripts
9. Write output



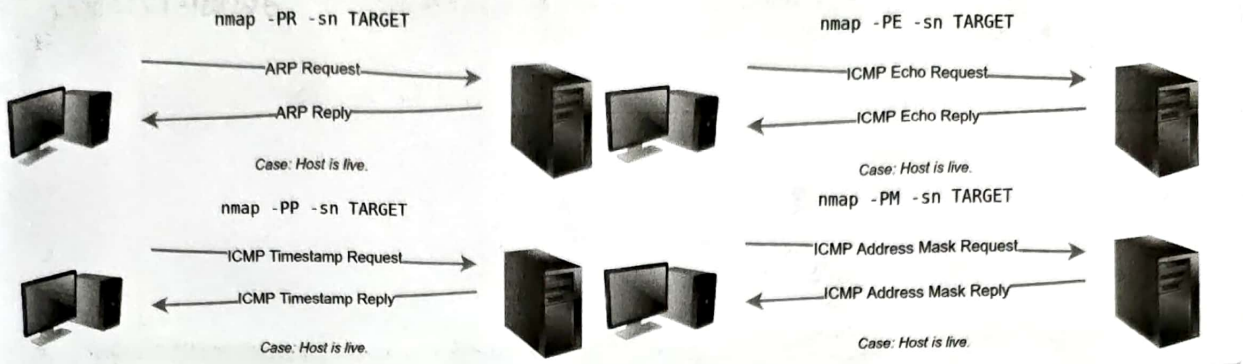


	ISO/OSI	TCP/IP	
7	Application Layer	Application Layer	HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP,...
6	Presentation Layer		
5	Session Layer		
4	Transport Layer	Transport Layer	TCP, UDP
3	Network Layer	Network Layer	IPv4, IPv6, ICMP, IPsec
2	Data Link Layer	Link Layer	ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth,
1	Physical Layer		





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Result :

Thus, the experiment was successfully completed. The different Nmap live host discovery techniques using ARP, ICMP, TCP and UDP were studied and used to identify active hosts.

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